

ASSESSMENT TOOL 1

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4

Total Marks:40

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.
4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding ; limited or partial integration	Poor understanding ; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Experiment 2

Configuration of IPv4 Addressing Between Two LANs and Analysis of IPv4 Header Fields Using Wireshark

Aim

To design a network topology consisting of two LANs connected through a router using Cisco Packet Tracer, configure IPv4 addressing, subnet masks, and default gateways, verify end-to-end connectivity using the ping command, and analyze the IPv4 header fields of the captured ICMP packets using Wireshark.

Procedure

1. Open Cisco Packet Tracer and place two switches, two PCs (one on each LAN), and a router with two Fast Ethernet interfaces.
2. Connect each PC to its respective switch, and connect each switch to a separate router interface (Fa0/0 and Fa0/1) using straight-through cables.
3. Assign a unique IPv4 address, subnet mask, and default gateway to each PC through the Desktop → IP Configuration option.
4. Configure the router interfaces with the corresponding IPv4 addresses and subnet masks, and enable each interface using the no shutdown command.
5. Verify end-to-end connectivity by pinging PC2 from PC1 (and vice versa) using the Command Prompt.
6. Switch to Simulation mode / attach Wireshark on the link between the router and a LAN, and capture the packets generated while pinging.
7. Open the captured ICMP packet in Wireshark and examine the IPv4 header fields — Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length.

Observation / Screenshots

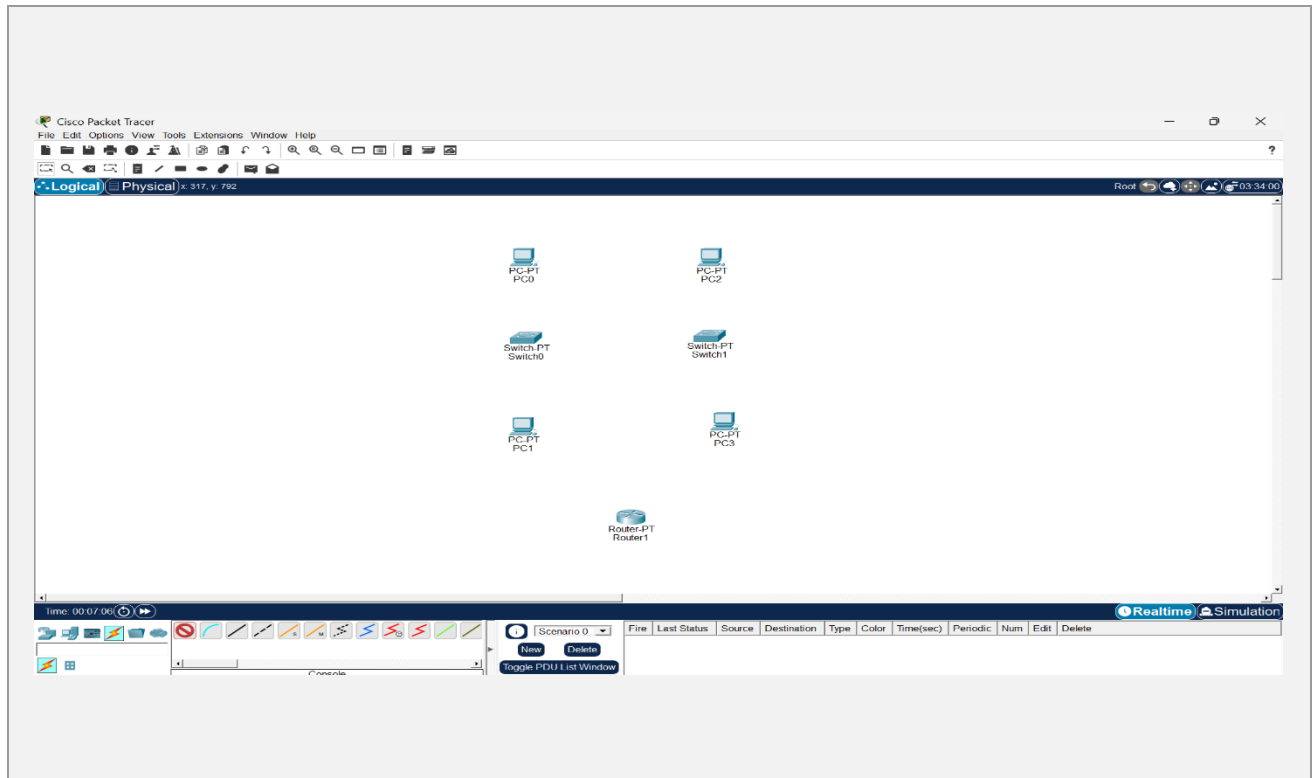


Fig 1: Network topology of two LANs connected through a router

EXPLANATION: Unconnected network hardware devices—four PCs (PC0 to PC3), two switches (Switch0 and Switch1), and a central router (Router1)—are placed onto the Cisco Packet Tracer workspace. This forms the initial, unlinked layout for a two-subnet network topology.

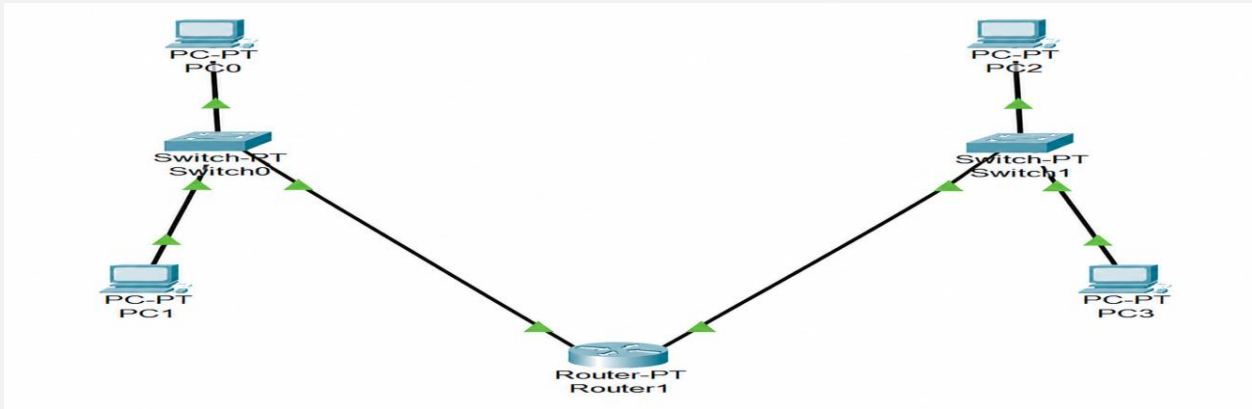


Fig 2: IPv4 address, subnet mask, and default gateway configuration on a PC

EXPLANATION: Straight-through Ethernet copper cables are wired to connect the end-user PCs to their respective switches, and each switch is linked directly to the central router. Green link indicators signal that the physical layer interfaces have established active link connections.

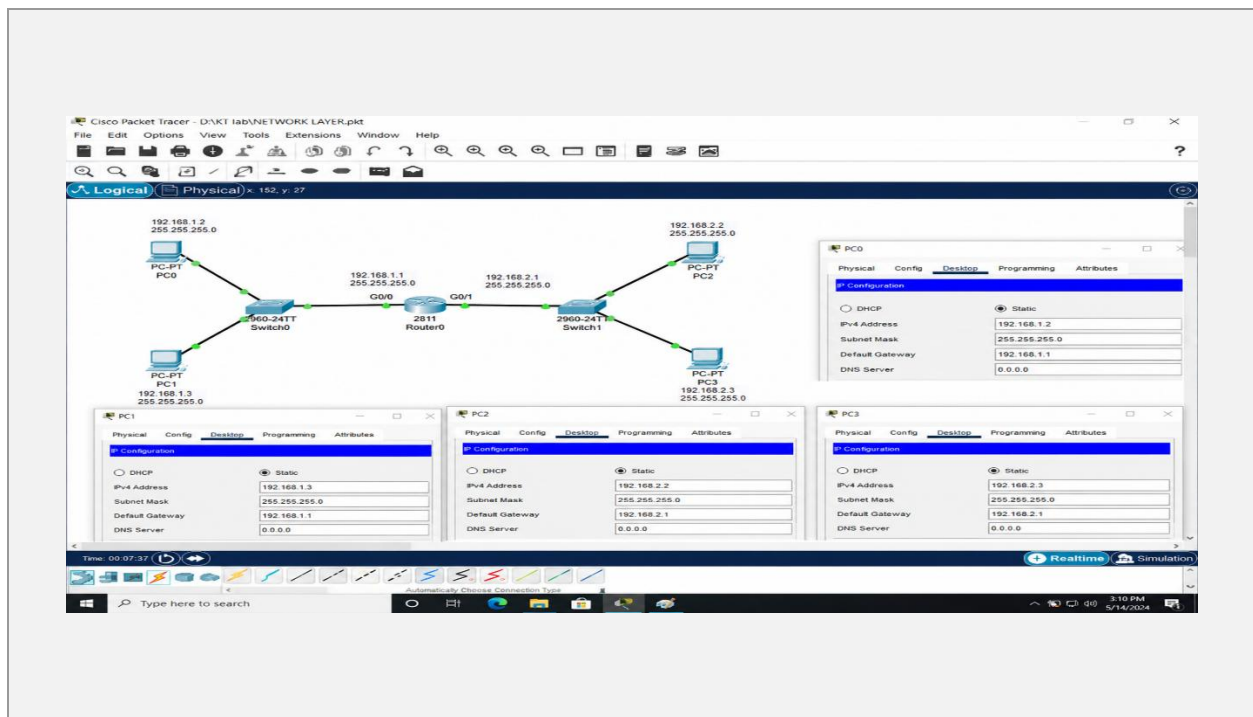


Fig 3: Successful ping showing end-to-end connectivity between the two LANs

EXPLANATION: Static IPv4 configuration windows are opened across the PCs to assign network parameters, including IP addresses, subnet masks, and default gateways (e.g., 192.168.1.1 and 192.168.2.1). This assigns PC0/PC1 to Subnet 1 and PC2/PC3 to Subnet 2 across the router's dual interfaces.

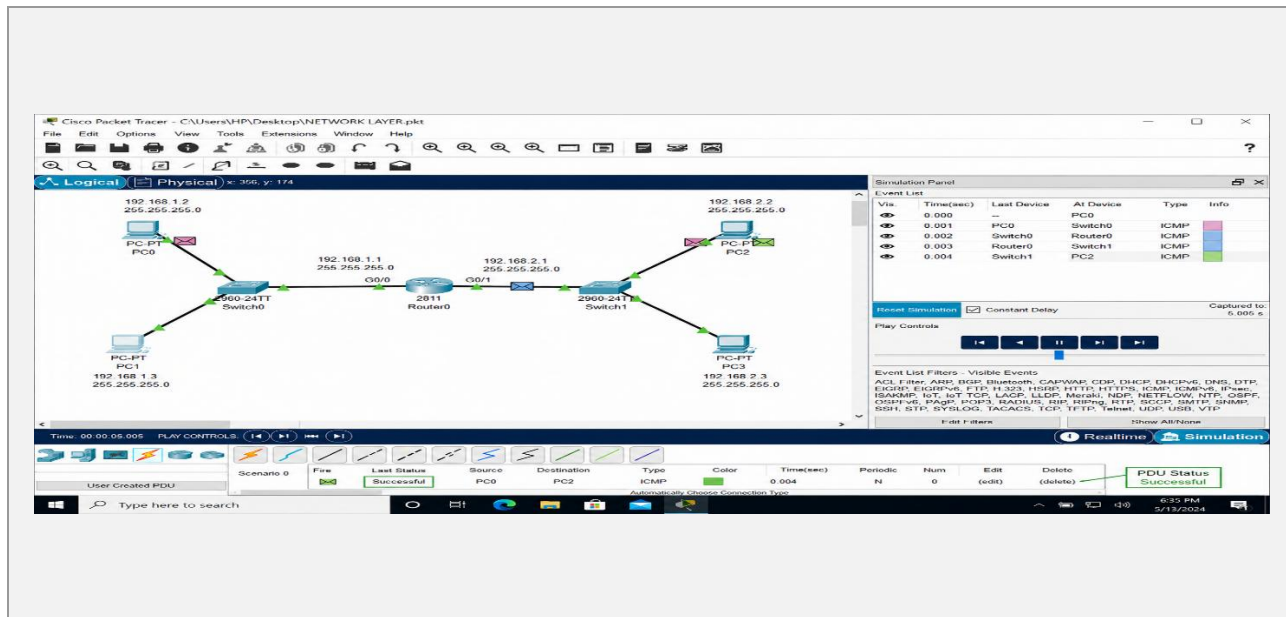


Fig 4: Wireshark capture showing the IPv4 header fields of the ICMP packet

EXPLANATION: A test ICMP packet (Simple PDU) is sent from PC0 to PC2 in Packet Tracer's Simulation Mode. The Event List logs packet hops across the router, and the status bar displays Successful, confirming inter-subnet communication.

The Wireshark capture shows a series of ICMP Echo (ping) requests and replies. The source IP is 10.158.252.31 and the destination IP is 8.8.8.8. The packets are captured on the Ethernet II interface.

No.	Time	Source	Destination	Protocol	Length	Info
1004	37.685927800	10.158.252.31	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply)
1057	38.641691100	10.158.252.31	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply)
1058	38.676087800	8.8.8.8	10.158.252.31	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=128 (request)
1088	39.657355600	10.158.252.31	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply)
1089	39.702461500	8.8.8.8	10.158.252.31	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=128 (request)
1103	40.670536600	10.158.252.31	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply)
1104	40.715661500	8.8.8.8	10.158.252.31	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=128 (request)

Frame 1003: Packet, 74 bytes on wire (592 bits), 74 bytes captured on interface CloudNetwork_1c:d3:83 (dc:56:7b:1c:d3:83), Dst: 8.8.8.8, Src: 10.158.252.31, Internet Protocol Version 4, Src: 10.158.252.31, Dst: 8.8.8.8, Internet Control Message Protocol

Fig 5: Wireshark capture showing the IPv4 header fields of the ICMP packet

EXPLANATION:A live packet capture in Wireshark uses the icmp display filter to trace continuous traffic between host 10.158.252.31 and Google DNS (8.8.8.8). The top pane shows matching pairs of Echo Requests and Echo Replies along with sequence numbers and TTL values.

```

C:\Windows\system32\cmd.e  X  +  v
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\jesin>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=61ms TTL=112
Reply from 8.8.8.8: bytes=32 time=34ms TTL=112
Reply from 8.8.8.8: bytes=32 time=45ms TTL=112
Reply from 8.8.8.8: bytes=32 time=45ms TTL=112

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 34ms, Maximum = 61ms, Average = 46ms

C:\Users\jesin>|
  
```

Fig 6: Wireshark capture showing the IPv4 header fields of the ICMP packet

EXPLANATION: The Windows Command Prompt displays the execution of ping 8.8.8.8. It shows 4 transmitted ICMP Echo Request packets receiving 0% loss with round-trip times ranging between 34 ms and 61 ms (average 46 ms).

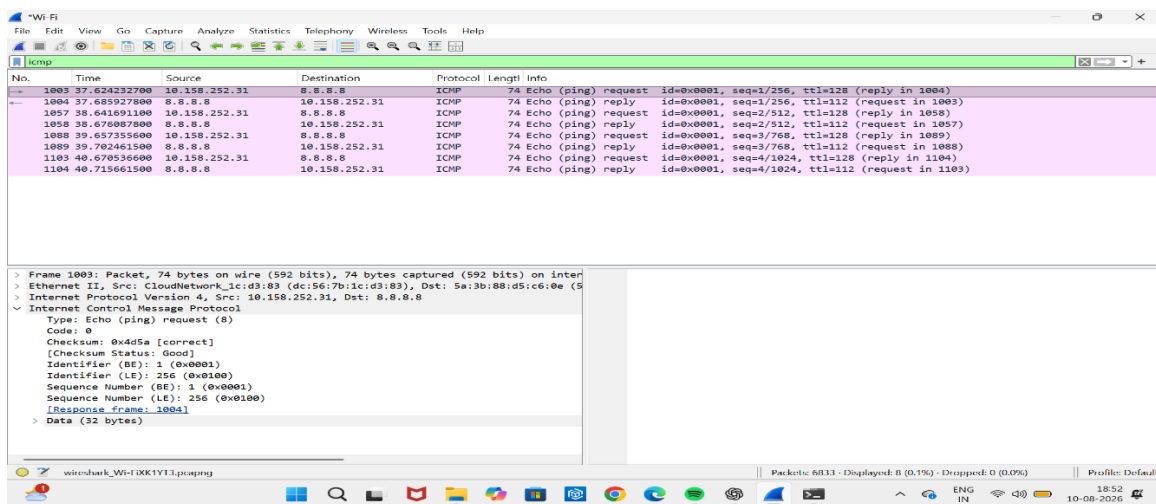


Fig 7: Wireshark capture showing the IPv4 header fields of the ICMP packet

EXPLANATION: Wireshark ICMP Header Inspection Wireshark expands the lower-layer Internet Control Message Protocol (ICMP) details for Frame 1003. It reveals the exact header fields of an Echo Request packet, showing Type: 8 (Echo request), Code: 0, valid checksum, Identifier, and Sequence values.

Result

The two LANs were successfully connected through a router, and IPv4 addressing, subnet masks, and default gateways were correctly configured on all devices. End-to-end connectivity was verified using the ping command, and the IPv4 header fields (Source IP, Destination IP, TTL, Protocol Number, and Header Length) were successfully captured and analyzed using Wireshark. Thus, the role of the Network layer in forwarding packets between different networks was studied and verified successfully.

Experiment 4

Analysis of ICMP Echo Request/Reply Messages and Round Trip Time (RTT) Calculation

Aim

To develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer, configure the required IP addresses, verify connectivity using the ping command, capture the ICMP traffic using Wireshark, identify the Echo Request and Echo Reply messages, examine the ICMP header fields, and calculate the Round Trip Time (RTT).

Procedure

1. Open Cisco Packet Tracer and place two PCs and one router; connect each PC to a router interface using an appropriate straight-through/copper cable.
2. Assign IPv4 addresses and subnet masks to both PCs, setting the router interface address of the corresponding subnet as the default gateway on each PC.
3. Configure the router's Fast Ethernet interfaces with the respective IPv4 addresses and enable them using the no shutdown command.
4. Verify that both interfaces show a green link light, confirming the physical and logical connections are active.
5. Enable Wireshark capture (or Simulation mode in Packet Tracer) on the link connecting the source PC to the router.
6. From the Command Prompt of the source PC, execute the ping command to send ICMP Echo Request messages to the destination PC.
7. In the captured trace, identify the ICMP Echo Request and Echo Reply pairs, examine the Type, Code, Identifier, and Sequence Number fields, and calculate the RTT as the time difference between each Echo Request and its corresponding Echo Reply.

Observation / Screenshots

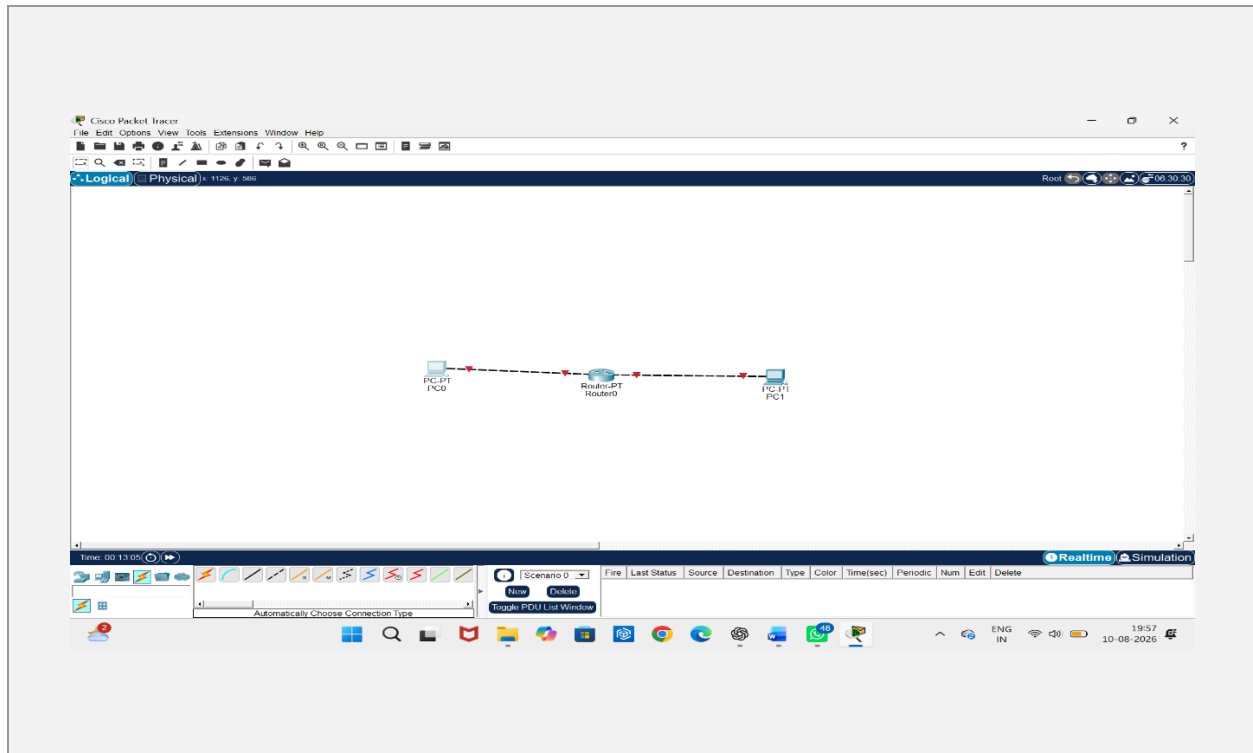


Fig 1: Network topology with two PCs connected through a router

EXPLANATION: Two PCs (PC0 and PC1) are connected to a central router (Router0) via serial/copper links on separate network interfaces. Red link indicators show that the router interfaces are shut down or unconfigured.

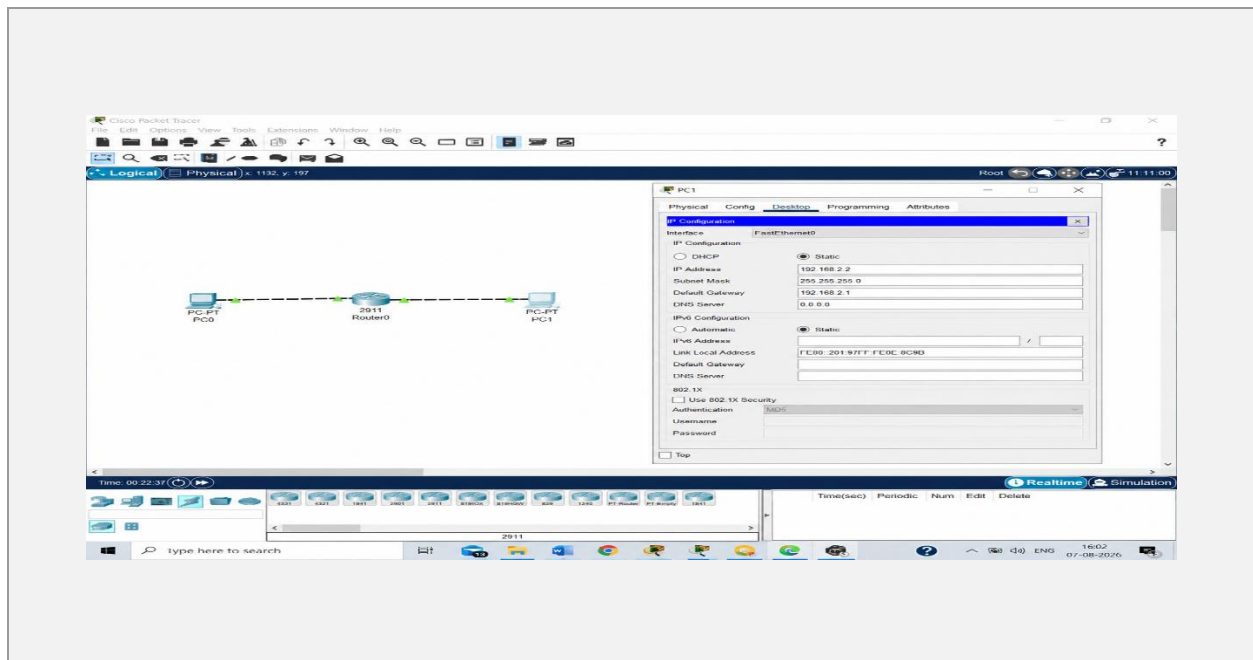


Fig 2: IP address and default gateway configuration on the source PC

EXPLANATION:The IP Configuration window for PC1 sets a static IP of 192.168.2.2 with subnet mask 255.255.255.0 and default gateway 192.168.2.1. Green indicators on the topology confirm the physical interfaces are up and active.

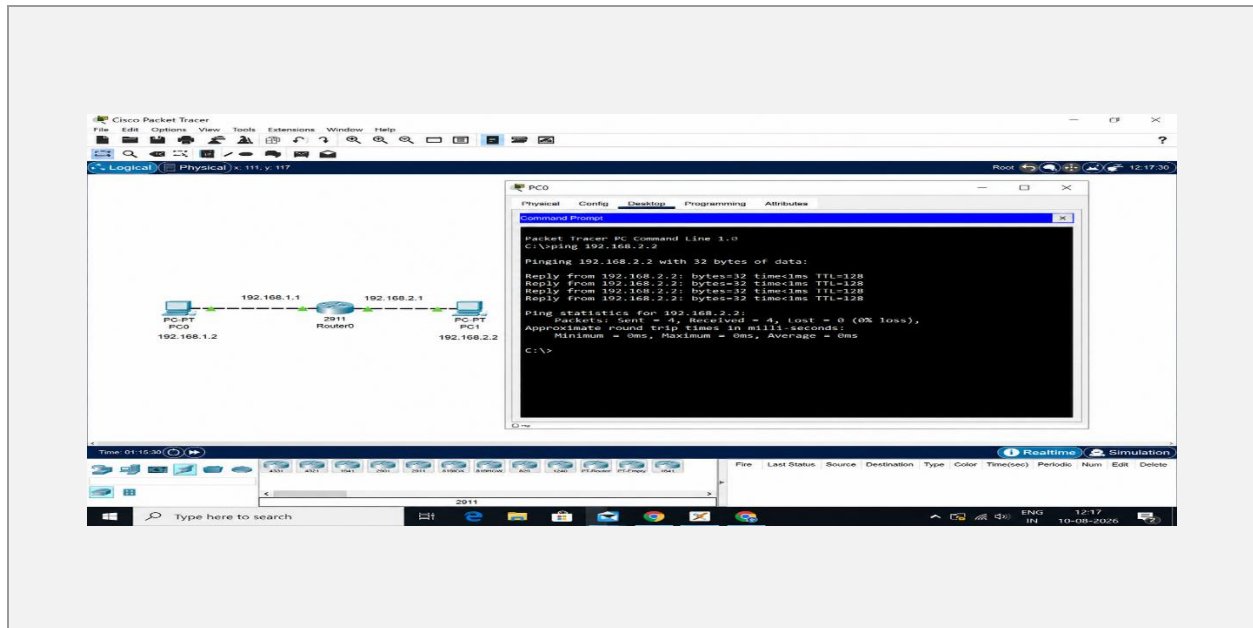


Fig 3: Ping output on the PC showing Echo Reply and RTT (time) values

EXPLANATION: The Command Prompt on PC0 runs a ping 192.168.2.2 command to reach PC1 across the router. Four successful reply packets are returned with 0% packet loss and round-trip times under 1 ms.

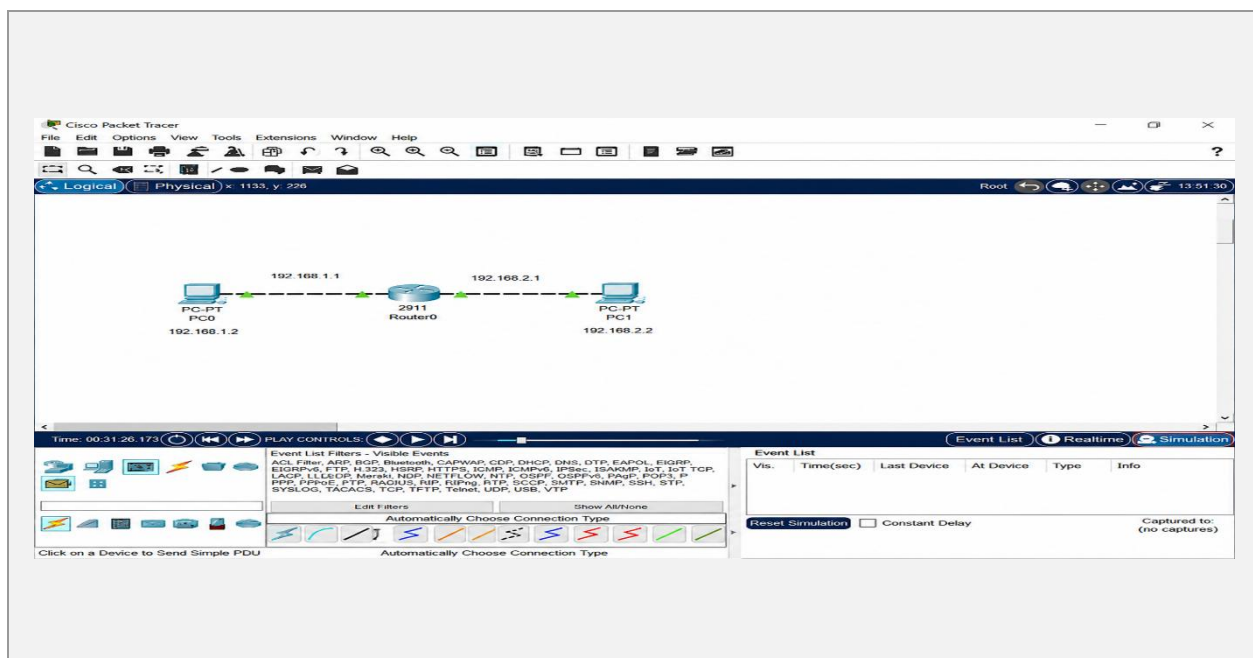


Fig 4: Wireshark capture showing ICMP Echo Request and Echo Reply with Type, Code, Identifier, and Sequence Number fields

EXPLANATION: Packet Tracer is toggled from Realtime mode to Simulation Mode to prepare for packet capture. The workspace shows PC0 (192.168.1.2), Router0, and PC1 (192.168.2.2) ready for simple PDU creation.

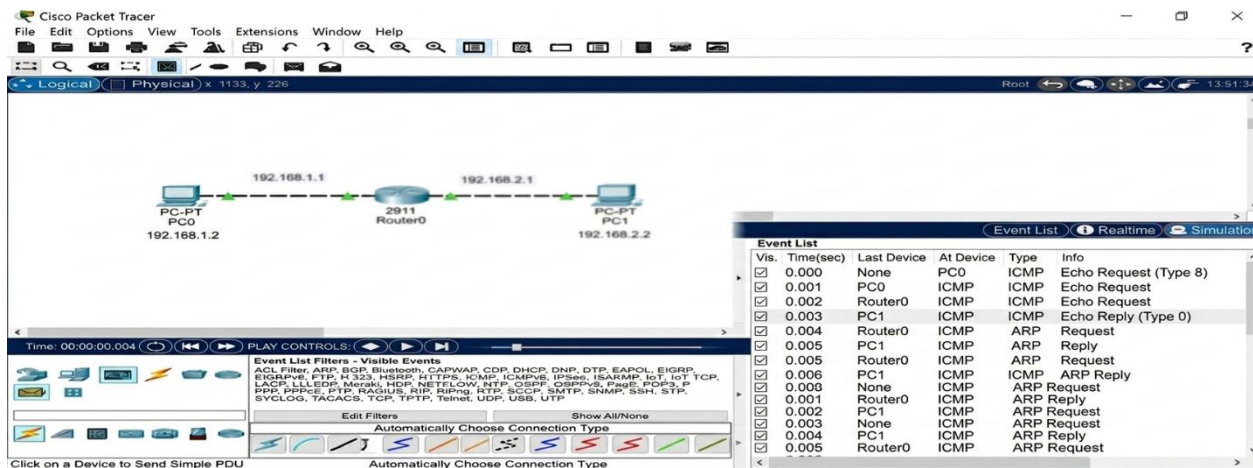


Fig 5: Wireshark capture showing ICMP Echo Request and Echo Reply with Type, Code, Identifier, and Sequence Number fields

EXPLANATION: A Simple PDU (ICMP packet) is sent from PC0 to PC1 using the envelope tool. The **Event List** on the bottom-right logs the generated ICMP Echo Requests (Type 8) and Echo Replies (Type 0) as they travel through the network.

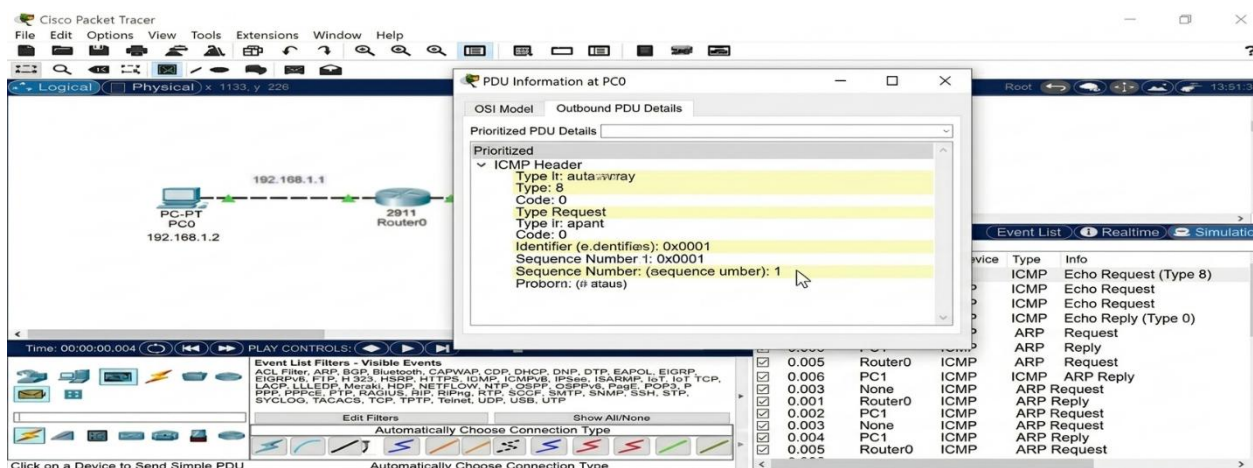


Fig 6: Wireshark capture showing ICMP Echo Request and Echo Reply with Type, Code, Identifier, and Sequence Number fields

EXPLANATION: Clicking the ICMP event opens the PDU Information window at PC0. The Outbound PDU Details tab reveals the ICMP header values for the Echo Request, showing Type: 8, Code: 0, and the matching Sequence/Identifier numbers.

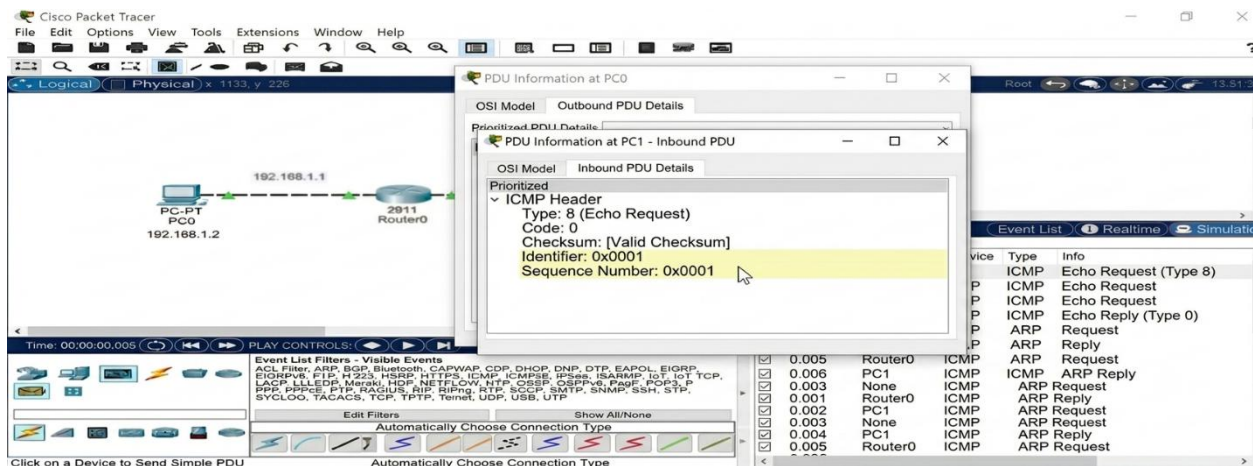


Fig 7: Wireshark capture showing ICMP Echo Request and Echo Reply with Type, Code, Identifier, and Sequence Number fields

EXPLANATION: The PDU Information window displays the Inbound PDU Details received at PC1. It confirms that the destination received a valid Echo Request (Type: 8, Code: 0) before generating its corresponding Echo Reply.

Result

The network with two PCs connected through a router was successfully configured and tested for connectivity using the ping command. The ICMP Echo Request and Echo Reply messages were captured using Wireshark, and header fields such as Type, Code, Identifier, and Sequence Number were identified. The Round Trip Time (RTT) was calculated from the timestamps of the request-reply pairs, confirming successful, low-latency communication. This experiment demonstrated how ICMP assists in verifying connectivity and troubleshooting network issues.